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Robot manipulator for eye surgery tool

Abstract

An eye surgery apparatus includes an eye surgery tool, an imaging system, a robotic arm, and a processor. The eye surgery tool has a distal end for insertion into an eye of a patient through an incision in the eye. The imaging system is configured to acquire images showing the incision and at least part of the eye surgery tool. The robotic arm is coupled with the eye surgery tool, which is configured to move the distal end of the eye surgery tool inside the eye according to one or more commands issued during an eye surgery. The processor is configured to, during the eye surgery (i) receive the images from the imaging system, (ii) monitor the commands issued to the robotic arm, (iii) detect, by analyzing the images, that a monitored command is expected to enlarge the incision, and (iv) initiate responsive action with respect to the detected command.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10744035	12/2019	Alvarez et al.	N/A	N/A
2014/0194859	12/2013	Ianchulev	N/A	N/A
2018/0353645	12/2017	Zhang	N/A	A61F 9/0008
2022/0079808	12/2021	Gliner et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2017100463	12/2016	WO	N/A

OTHER PUBLICATIONS

Bourcier T., et al., “Robot-Assisted Simulated Cataract Surgery,” Journal Cataract and Refractive Surgery, May 2017, vol. 43(4), pp. 552-557, XP085033601, ISSN: 0886-3350, DOI: 10. 1016/J.JCRS.2017.02.020. cited by applicant
Gerber, M.J., Pettenkofer, M. & Hubschman, JP. Advanced robotic surgical systems in ophthalmology. Eye 34, 1554-1562 (2020). https://doi.org/10.1038/s41433-020-0837-9. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application 63/297,398, filed Jan. 7, 2022, whose disclosure is incorporated herein by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates generally to robotic medical systems for eye surgery, and

particularly to robotic phacoemulsification systems.
BACKGROUND OF THE DISCLOSURE
(2) An eye surgery may be required in some ophthalmic medical conditions. For example, a physician may recommend cataract removal using phacoemulsification cataract surgery. In the procedure, the surgeon makes a small incision in the sclera or cornea of the eye. Then a portion of the anterior surface of the lens capsule is removed to gain access to the cataract. The surgeon then uses a phacoemulsification probe, which has an ultrasonic handpiece with a needle. The tip of the needle vibrates at ultrasonic frequency to sculpt and emulsify the cataract while a pump aspirates particles and fluid from the eye through the tip. Aspirated fluids are replaced with irrigation of a balanced salt solution to maintain the anterior chamber of the eye. After removing the cataract with phacoemulsification, the softer outer lens cortex is removed with suction. An intraocular lens (IOL) is then introduced into the empty lens capsule restoring the patient's vision.
(3) Various techniques to invasively treat a cataracted eye were proposed in the patent literature. For example, U.S. Pat. No. 10,744,035 describes systems and processes for facilitating the removal of cataract material with a robotically assisted tool with laser, irrigation capabilities, aspiration capabilities. Tool guidance systems that make use of vision technologies, including optical coherence tomography (OCT), white light imaging, and structured light imaging. Emulsification patterns optimized to minimize risk to the patient and reduce procedure time. Robotic tools with articulation capabilities that allow for precise control during capsulorhexis and emulsification procedures. Robotic instrument drive mechanisms combined with pumps, flow meters, and valves regulate and control irrigation and aspiration functionalities during robotic ophthalmologic procedures. Use of a robotically controlled articulating tip may minimize the size of the incision in the lens capsule necessary to extract the cataract material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic, pictorial view of a phacoemulsification apparatus comprising a phacoemulsification probe coupled to a robotic arm for moving the phacoemulsification probe inside an eye using an incision as a pivot point, in accordance with an example of the present disclosure;
(2) FIG. **2** is a block diagram schematically illustrating the operation of the apparatus of FIG. **1** using an incision as a pivot point for the phacoemulsification probe, in accordance with an example of the present disclosure; and
(3) FIG. **3** is a flow chart schematically illustrating an incision pivot point operating method of the apparatus of FIG. **1**, in accordance with an example of the present disclosure.
DETAILED DESCRIPTION OF EXAMPLES
Overview
(4) Eye surgery may involve inserting a number of surgical tools into the eye via small incisions made in the surface of the eye. For example, cataract surgery typically requires two incisions in the cornea: one for a phacoemulsification probe, and a second for a different tool such as a curette. The surgeon performing the procedure adjusts movements of the inserted tools in such a way so as not to enlarge the incisions. However, if the tools are operated robotically, there may be no such limitation placed on the robots.
(5) Examples of the present disclosure that are described hereinafter provide a robotic eye surgery apparatus and algorithms to perform eye surgery using tools held by robotic arms of the apparatus while maintaining minimal and stable incision entry points during the procedure.
(6) In some examples, the processor identifies the position of the eye surgery tool relative to the incision, by analyzing images acquired in real-time. Based on the identified position, the processor

detects that the command is expected to enlarge the incision. To this end, the coordinates of any incision are identified by image processing (e.g., using edge detection) and the processor can determine if a requested track of the tip (given in a same coordinate system) will comprise the incision.
(7) Typically, a processor of the apparatus instructs the robots to move the distal tips in response to commands the processor receives from a surgeon performing the surgery, which the processor finds will not increase the incision. The command can be alternatively received from a robot that performs the surgery automatically using a suitable algorithm, and in such a case the processor verifies the robot automatic operation does not increase the incision. Alternatively, a treatment plan (e.g., an order of removal of a cataract) is uploaded to the processor so it moves the distal tips (e.g., tips of rigid distal ends) so the distal tips cover a volume of the lens according to the plan. The processor overrides any step in the treatment plan that may increase the incision. For both robotic methods, the means described below ensure that the distal end motions do not cause unwanted enlargement of the incisions.
(8) In one example, two robotic arms, one for the phacoemulsification probe, the other for a second tool, are controlled by the processor, which also controls an imaging system. The imaging system images the incisions made in the eye and the distal tips of the phacoemulsification probe and the tool inside the eye. From the images, the processor determines the locations of the incisions and the distal tips. The processor uses this information to calculate robotic motion of the distal ends (that can be fully rigid, or sufficiently rigid relative to eye tissue) that is safe to tissue near the incision, as further described below.
(9) In an example, before performing any given movement, for example moving the distal tip of the phacoemulsification probe several millimeters in a direction to emulsify the lens, the processor calculates how the requested movement can be made so that the only motion of the phacoemulsification probe at the incision is rotation around the incision, using the incision as a pivot point of the rigid distal end, and/or translation of the distal end into or out of the incision. These types of movements do not enlarge the incision, and so the processor calculates and commands the robotic arm (that has a sufficient number of degrees of freedom) to make only these types of movements.
(10) In another example, the processor overrides any command made by a surgeon that is predicted by the processor calculations to enlarge an incision. Namely, no movement of the rigid distal ends which might enlarge an incision is permitted. In another example, if the surgeon further requests such a movement, the processor warns the surgeon that the request may enlarge the incision.
(11) Using the disclosed apparatuses and methods for robotic eye surgery (e.g., emulsification) may allow more accurate and less hazardous eye surgeries, such as cataract surgeries.
APPARATUS DESCRIPTION
(12) FIG. **1** is a schematic, pictorial view of an eye surgery apparatus **10** comprising robot-mounted eye surgical tools **12** and **13**, in accordance with an example of the present disclosure. Surgical tools **12** and **13** are mounted on robot arms **44** and **66**, respectively, and have rigid distal ends **14** and **17**, respectively, engaging a lens capsule **18** of an eye **20** of a patient **19**. The rigid distal ends are inserted via respective incisions **8** and **9** in the cornea of eye **20**, the incisions made just before by another tool, e.g., by a physician **11**.
(13) In the example shown in inset **25**, rigid distal end **14** comprises, by way of example, phacoemulsification probe **12**, which itself comprises an aspiration channel **15** and an irrigation channel **16** (which can be coaxial or side-by-side). As shown, aspiration channel **15** has an inlet **115**, and irrigation channel **16** has an outlet **116**, both at a distal tip of phacoemulsification probe **12**, from which cataract fragments are aspirated and into which replenishing irrigation fluid flows, respectively. Rigid distal end **14** is shown straight, though it may be curved or bent. Rigid distal end **17** is of another surgical tool (**13**), for example a curette.
(14) Robotic arms **44** and **66** (mounted on a base **102**) are configured to, at minimum, tilt directions

of respective distal ends **14** and **17** within lens capsule **18**, and adjust a depth of the distal ends inside the eye, while maintaining axes, or pivots, of distal ends **14** and **17** at respective incisions **8** and **9** in eye **20**, so as not to enlarge incisions **8** and **9**, as described in FIG. **2**. While the shown example uses robotic arms **44** and **66**, each having six degrees of freedom, the number of degrees of freedom may vary with design, to maintain the aforementioned axes at incisions **8** and **9** while the distal tips of rigid distal ends **14** and **17** move inside eye **20**.

(15) In the exemplified example, during the phacoemulsification procedure robotic arm **44** moves rigid distal end **14** so that aspiration inlet **115** and irrigation outlet **116** follow a path (shown in FIG. **2**) inside the eye, according to, for example, commands from a processor **38** communicated via a cable **31**, that implements a treatment plan (e.g., one stored in a memory **35**) made by physician **11**. Similarly, robotic arm **66** moves rigid distal end **17** according to commands from processor **38** communicated via a cable **33**, typically in a coordinated track with that of rigid distal end **17**.

(16) In the shown example, console **28** comprises a piezoelectric drive module **30**, which is coupled, using electrical wiring running in cable **37**, with a piezoelectric crystal inside probe **12** that generates vibration of rigid distal end **14** that breaks cataracted lens **18**. Drive module **30** is controlled by processor **38** to adjust a vibration power and/or duration and/or frequency.

(17) During the phacoemulsification procedure, a pumping subsystem **24** comprised in a console **28** pumps irrigation fluid from an irrigation reservoir to outlet **116** to irrigate the eye. In an example, the irrigation fluid may be administered via a gravity-fed method or any other known method in the art. The fluid is pumped via a tubing line **43** running from the console **28** to phacoemulsification probe **12**. Eye fluid and waste matter (e.g., emulsified parts of the cataract) are aspirated via inlet **115** to a collection receptacle (not shown) by a pumping subsystem **26** also comprised in console **28** and using a tubing line **46** running from phacoemulsification probe **12** to console **28**.

(18) In the shown example, apparatus **10** further comprises an eye imaging system **58** mounted on base **102**, e.g., a video camera with an optical imaging axis **155**, the camera capturing an image (e.g., records video images) of the aforementioned incisions **8** and **9**, and of lens capsule **18**, both in real time, including the inserted rigid distal ends. The captured image **69** is displayed on a display **36**. Real time image processing enables processor **38** to calculate the required motions of robotic arms **44** and **66**, so as to perform the eye surgery without enlarging eye incisions **8** and **9**, or to override surgeon's **11** commands that may cause enlargement of an incision, as described below.

(19) Processor **38** presents other results of the cataract removal procedure on display **36**. Processor **38** may receive user-based commands via a user interface **40**, which may include setting or adjusting an irrigation rate and/or aspiration rate. User interface **40** may be combined with a touch screen graphical user interface of display **36**.

(20) Some or all of the functions of processor **38** may be combined in a single physical component or, alternatively, implemented using multiple physical components. These physical components may comprise hard-wired or programmable devices, or a combination of the two. In some examples, at least some of the functions of processor **38** may be carried out by suitable software stored in memory **35** (as shown in FIG. **1**). This software may be downloaded to a device in electronic form, over a network, for example. Alternatively, or additionally, the software may be stored in tangible, non-transitory computer-readable storage media, such as optical, magnetic, or electronic memory.

(21) The apparatus shown in FIG. **1** may include further elements, which are omitted for clarity of presentation. For example, physician **11** may hold a control handle with which the physician can, for example, abort the automatic procedure. Physician **11** may apply medications with another tool, which is also not shown in order to maintain clarity and simplicity of presentation.

(22) While the exemplified surgical procedure involves corneal incisions, incisions may be performed at other eye locations, depending on the type of eye surgery. Enlarging incisions in such other locations (e.g., at the sclera) would similarly need to be minimized during the surgery, to

prevent damage to the eye. While FIG. **1** refers to eye surgical tools, the technique can also be used with eye diagnostic devices (e.g., a miniature camera, illuminator, pressure and/or temperature sensors, among others) fitted at a distal portion of a rigid distal end for insertion into the eye.

Robot Manipulators for Eye Tools

(23) FIG. **2** is a block diagram schematically illustrating the operation of apparatus **10** of FIG. **1** using incision **8** as a pivot point for phacoemulsification probe **12**, in accordance with an example of the present disclosure.

(24) As seen, rigid distal ends **14** and **17** of respective probe **12** and tool **13** are located partially inside eye **20**, accessing the eye via incisions **8** and **9**. Probe **12** and tool **13** are coupled to respective robotic arms **44** and **66** to perform an operation inside lens capsule **18**, such as phacoemulsification (using probe **12**) and curetting (using tool **13**). To this end, by way of example, a distal portion **111** of distal rigid end **14** is inside the eye, with portion **111** length measured between a pivot point **250** at incision **8** and a tip **112** of rigid distal end **14**.

(25) Assuming tip **112** is required to follow a path **222**, processor **38** calculates the movement required by robotic arm **44** and instructs the robotic arm to move tip **112** along path **222** by changing only a pivot angle **114** and length of distal portion **111** within the eye. In this way, incision **8** is not enlarged by the required motion. As illustrated by layout **212**, the robotic arm moves the probe (now designated **12′**) to a new position and orientation such that rigid distal end **14** goes through the same pivot point **250** to a new location over path **222** (i.e., by having a different angle and portion length).

(26) Processor **38** calculates the required motion using information from imaging system **58** that can be a video camera with an optical imaging axis **155** and field of view (FOV) **255**. A video camera of imaging system **58** captures at least part of the eye, including incisions **8** and **9** and the distal tips (e.g., tip **112**) of the phacoemulsification probe and the curetting tool, and from these images the processor determines if and how to move the distal tips.

(27) The example apparatus shown in FIG. **2** is chosen purely for the sake of conceptual clarity. For example, the rigid distal ends may be curved or bent. As another example, imaging system **58** may include built-in image processing circuitries to perform the required image processing.

(28) FIG. **3** is a flow chart schematically illustrating an incision pivot point operating method of apparatus **10** of FIG. **1**, in accordance with an example of the present disclosure. The algorithm, according to the presented example, carries out a process that begins with physician **11** operating apparatus **10** (e.g., processor **38**) to command robotic arm **44** to have a surgical tool, such as diamond tipped knife perform the incision (e.g., incision **8**) in eye **20**, at an eye incision step **302**. Alternatively, the incision may be performed manually by physician **11**, using, for example, a scalpel or another type of blade.

(29) At a distal end insertion step **304**, processor **38** commands robotic arm **44** to have tool **12** insert a distal portion **111** of rigid distal end **14** through incision **6** into a predetermined location inside eye **20**.

(30) At imaging step **306**, imaging system **58** images incision **8** and tip **112** of rigid distal end **14**.

(31) Processor **38** receives the images and perform image processing to determine coordinates of incision **8** and tip **112** of rigid distal end **14** inside eye **20**, at an image processing step **308**.

(32) The processor receives a command from surgeon **11** at a command receiving step **309**, to move rigid distal end **14** inside eye **20** to, for example, perform phacoemulsification at a new location in lens **18**.

(33) Processor **38** monitors the commands issued by the surgeon to the robotic arm, to check if a monitored command is expected to enlarge the incision. To this end, processor **38** analyzes the images processed in step **308**. Using the coordinates determined in step **308**, processor **38** calculates the required motion of rigid distal end **14** and checks if the motion will enlarge incision **8**, at a checking step **310**.

(34) If the answer is yes, processor **38** initiates a responsive action with respect to the monitored

and detected potentially hazardous tip moving command, e.g., responsive action such as overriding the command and providing an audio and/or visual warning to physician **11**, at a command overriding step **312**.
(35) In some examples, the processor is configured to block the detected command from being executed by the robotic arm and/or to output an alert to the surgeon with respect to the detected command.
(36) If the answer is no, processor **38** instructs robotic arm **44** to move rigid distal end **14** inside eye **20** over a path as required by the commanded step (e.g., to follow track **222**) while maintaining incision **8** as a pivot point of the motion of the distal end, at tool operation step **314**.
(37) The example flow chart shown in FIG. **3** is chosen purely for the sake of conceptual clarity. For example, as described above, additional tools may be inserted via different incisions in the eye.
Example 1
(38) An eye surgery apparatus (**10**), comprising (a) an eye surgery tool (**12**, **13**) having a distal end (**14**, **17**) for insertion into an eye (**20**) of a patient (**19**) through an incision (**8**, **9**) in the eye, (b) an imaging system (**58**), which is configured to acquire images (**69**) showing the incision (**8**, **9**) and at least part of the eye surgery tool (**12**, **13**), (c) a robotic arm (**44**, **66**) coupled with the eye surgery tool (**12**, **13**), which is configured to move the distal end (**14**, **17**) of the eye surgery tool (**12**, **13**) inside the eye (**20**) according to one or more commands issued during an eye surgery, and (d) a processor (**38**), which is configured to, during the eye surgery (i) receive the images (**69**) from the imaging system (**58**), (ii) monitor the commands issued to the robotic arm (**44**, **66**), (iii) detect, by analyzing the images (**69**), that a monitored command is expected to enlarge the incision (**8**, **9**), and (iv) initiate a responsive action with respect to the detected command.
Example 2
(39) The eye surgery apparatus according to example 1, wherein the one or more commands are issued by one of a surgeon and an algorithm that the robotic arm (**44**, **66**) executes.
Example 3
(40) The eye surgery apparatus according to example 1 or 2, wherein the processor (**38**) is configured to identify a position of the at least part of the eye surgery tool (**12**, **13**), relative to the incision (**8**, **9**), by analyzing the images (**69**), and to detect, based on the identified position, that the command is expected to enlarge the incision (**8**, **9**).
Example 4
(41) The eye surgery apparatus according to example 1 or 2, wherein the processor (**38**) is configured to block the detected command from being executed by the robotic arm (**44**, **66**).
Example 5
(42) The eye surgery apparatus according to example 1 or 2, wherein the processor (**38**) is configured to output an alert to a surgeon with respect to the detected command.
Example 6
(43) The eye surgery apparatus according to any one of examples 1-5, wherein the eye surgery is a phacoemulsification procedure, the eye surgery tool is a phacoemulsification probe (**12**), and the imaging system (**58**) is configured to image a tip (**112**) of the phacoemulsification probe (**12**).
Example 7
(44) The eye surgery apparatus according to any one of examples 1-7, wherein the imaging system (**58**) comprises a video camera, and wherein the images (**69**) comprise video images.
Example 8
(45) An eye surgery method, comprising (a) inserting a distal end of an eye surgery tool into an eye of a patient through an incision in the eye, (b) using an imaging system, acquiring images showing the incision and at least part of the eye surgery tool, (c) using a robotic arm, moving the distal end of the eye surgery tool inside the eye according to one or more commands issued to the robotic arm, (d) receiving the images from the imaging system, (e) monitoring the commands issued to the robotic arm, (f) detecting, by analyzing the images, that a monitored command is expected to

enlarge the incision, and (g) initiating a responsive action with respect to the detected command.
(46) Although the examples described herein mainly address eye surgeries, and cataract surgeries in particular, the methods described herein can also be used in other minimally invasive surgical applications at other body locations (e.g., via skin or skull) using rigid tools having access via small incisions that require protection from enlargement.
(47) It will be thus appreciated that the examples described above are cited by way of example, and that the present disclosure is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and sub-combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art. Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that to the extent any terms are defined in these incorporated documents in a manner that conflicts with the definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

Claims

1. An eye surgery apparatus, comprising: an eye surgery tool having a distal end for insertion into an eye of a patient through an incision in the eye; an imaging system, which is configured to acquire images showing the incision and at least part of the eye surgery tool; a robotic arm coupled with the eye surgery tool, which is configured to move the distal end of the eye surgery tool inside the eye according to one or more commands issued during an eye surgery; and a processor, which is configured to, during the eye surgery: receive the images from the imaging system; monitor the commands issued to the robotic arm; detect, by analyzing the images, that a monitored command is expected to enlarge the incision; and initiate a responsive action with respect to the detected command.
2. The eye surgery apparatus according to claim 1, wherein the one or more commands are issued by one of a surgeon and an algorithm that the robotic arm executes.
3. The eye surgery apparatus according to claim 1, wherein the processor is configured to identify a position of the at least part of the eye surgery tool, relative to the incision, by analyzing the images, and to detect, based on the identified position, that the command is expected to enlarge the incision.
4. The eye surgery apparatus according to claim 1, wherein the processor is configured to block the detected command from being executed by the robotic arm.
5. The eye surgery apparatus according to claim 1, wherein the processor is configured to output an alert to a surgeon with respect to the detected command.
6. The eye surgery apparatus according to claim 1, wherein the eye surgery is a phacoemulsification procedure, the eye surgery tool is a phacoemulsification probe, and the imaging system is configured to image a tip of the phacoemulsification probe.
7. The eye surgery apparatus according to claim 1, wherein the imaging system comprises a video camera, and wherein the images comprise video images.
8. An eye surgery method, comprising: inserting a distal end of an eye surgery tool into an eye of a patient through an incision in the eye; using an imaging system, acquiring images showing the incision and at least part of the eye surgery tool; using a robotic arm, moving the distal end of the eye surgery tool inside the eye according to one or more commands issued to the robotic arm; receiving the images from the imaging system; monitoring the commands issued to the robotic arm; detecting, by analyzing the images, that a monitored command is expected to enlarge the incision; and initiating a responsive action with respect to the detected command.
9. The eye surgery method according to claim 8, wherein the one or more of commands are issued by one of a surgeon and an algorithm that the robotic arm executes.

10. The eye surgery method according to claim 8, wherein detecting that the monitored command is expected to enlarge the incision comprises identifying a position of the at least part of the eye surgery tool, relative to the incision, by analyzing the images, and detecting, based on the identified positions, that the command is expected to enlarge the incision.

11. The eye surgery method according to claim 8, wherein initiating the responsive action comprises blocking the detected command from being executed by the robotic arm.

12. The eye surgery method according to claim 8, wherein initiating the responsive action comprises outputting an alert to the surgeon with respect to the detected command.

13. The eye surgery method according to claim 8, wherein the eye surgery is a phacoemulsification procedure, the eye surgery tool is a phacoemulsification probe, and the imaging system is configured to image a tip of the phacoemulsification probe.

14. The eye surgery method according to claim 8, wherein the imaging system comprises a video camera, and wherein the images comprise video images.